The Lucas Critique Examined in ML

D200 Machine Learning in Economics

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Outline

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The Lucas Critique (1976)

Core Argument

Parameters of reduced-form econometric models are not policy-invariant. When a policy changes, agents update their expectations, altering the model's coefficients and rendering pre-change forecasts invalid.

Formally, if an agent's rule is $x_t = f(y_t; \theta)$ where θ embeds beliefs about policy $\{\pi_t\}$, then a shift $\pi \to \pi'$ forces $\theta \to \theta'$ — invalidating any forecast built on $\hat{\theta}$.

Classical response: VARs, regime-switching models — reduce theory dependence while retaining parametric structure.

ML: Model-agnostic, data-driven, flexible functional form. Do ML models provide a hedge against the Lucas Critique, or does over-fitting to the pre-break regime make things *worse*?

Research Question

Are regime-switching ML models more or less susceptible to the Lucas Critique than classical parametric alternatives, and **why**?

Paper's Headline

ML models degrade *similarly* to classical models but still produce **better out-of-sample forecasts** under regime change.

Literature Review

Classical response to Lucas:

- **VARs** (as cited in Bormetti & Corsi, 2022) — reduced-form, model-agnostic, no structural identification.
- **TVP-VARs** (Primiceri, 2005) — time-varying parameters; still parametric.
- **MSM / Hamilton filter** (1989) — latent regimes; parameters vary by regime, not by policy change.
- **DSGEs** — micro-founded; Storm (2021) argues still vulnerable to structural misspecification.

ML and the Lucas Critique:

- **Athey (2018)** — data-driven model selection; may outperform parametric models when structure shifts.
- **Chen (2018)** — ML on Progresa programme; relative forecasting strength under changing structure.
- **Salle (2015)** — ANN expectation formation; Lucas-robust as no structural form assumed.
- **Pereira (2025)** — ML rethinks econometric methodology (predictive economics manifesto).

Gap This Project Fills

No systematic *head-to-head* comparison of classical and ML regime-switching models on the same Lucas Critique experiment with both simulated (known DGP) and real macroeconomic data.

Theoretical Framework: Forecast Risk Decomposition

Suppose the true data-generating process is regime-dependent: $y_{t+1} = f_{s_t}(x_t) + \varepsilon_{t+1}, s_t \in \{1, \ldots, S\}$, where x_t are observed macro states and s_t is a latent regime (e.g. a policy framework). Structural change implies that the mapping $f_{s_t}(\cdot)$ varies across regimes, hence, the MSE-optimal forecast is

$$m^*(x_t) = \mathbb{E}[y_{t+1} \mid x_t] = \sum_{s=1}^{S} \Pr(s_t = s \mid x_t)\, \mathbb{E}[y_{t+1} \mid x_t, s_t = s].$$

Thus forecasting requires learning both regime probabilities and within-regime conditional means. For any forecasting model $\hat{m}(x)$, expected squared forecast risk satisfies $\mathbb{E}\big[(y_{t+1} - \hat{m}(x_t))^2\big]$

$$= \sigma^2 + \underbrace{\mathbb{E}\big[(m^*(x_t) - m(x_t; \mathcal{F}))^2\big]}_{\text{approximation error}} + \underbrace{\mathbb{E}\big[(m(x_t; \mathcal{F}) - \hat{m}(x_t))^2\big]}_{\text{estimation error}},$$

where $\mathcal{F}$ denotes the model class and $m(x_t; \mathcal{F})$ is the best model in the class. Flexible regime-switching ML models approximate the mixture directly:

$$\hat{m}_{\text{ML}}(x_t) = \sum_{s=1}^{S} \hat{g}_s(x_t)\, \hat{\mu}_s(x_t),$$

allowing state-dependent regime probabilities $\hat{g}_s(\cdot)$ and nonlinear conditional means $\hat{\mu}_s(\cdot)$.

Theoretical Framework: Insights

- ML Gains: by flexibly estimating both probabilities of being in a state (non-linear state estimation, $\hat{g}_s$) alongside functional estimation at each state of the mean, $\hat{\mu}_s$
- Parametric MSM: low estimation error (correct spec.) but highs *approximation* error once DGP shifts.
- Flexible ML: low approximation error in-sample but high *estimation* error post-break (over-fitting).
- Variance-reducing break (INDPRO 1984): σ^2 falls $\Rightarrow$ LSR < 1 even with frozen $\hat{\theta}$ — *inverse Lucas Critique*.
- Ensembles expand $\mathcal{F}$ via model averaging reduces approximation error — theoretical basis for the ensemble's stability.

Data: Two Experimental Settings

Experiment A — Simulated DGP

Two-state Markov-switching AR(1) with known ground truth:

$$y_t = \mu_{s_t} + \phi_{s_t} y_{t-1} + \sigma_{s_t} \varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

	Mild		Severe	
Param	Pre	Post	Pre	Post
μ_0	0.5	0.2	0.5	−0.5
μ_1	−1.5	−1.0	−1.5	−3.0
σ_0	1.0	1.3	1.0	2.0
σ_1	2.5	3.0	2.5	4.5
P_{00}	0.95	0.90	0.95	0.80

Mild = gradual tightening; Severe = Volcker-scale

Experiment B — Real-world FRED data

Series	Break	Period
INDPRO	Great Moderation	Jan 1984
CPIAUCSL	Volcker disinflation	Oct 1979

Covariates: EFFR, UNRATE, USREC (NBER).

Transformations: log-differences ×100; lags; 5/20-period rolling mean/std; trend-cycle decomposition for HMM features.

Protocol: Train pre-break only; evaluate post-break. No re-training or look-ahead. Models frozen at pre-break estimates.

Eight Models: A Taxonomy

Model	Family	Regime identification	Forecast
AR(2)	Linear baseline	None	AR linear
ARMA(2,1)	Linear baseline	None	AR+MA linear
MSM	Classical reg.-switch	Hamilton filter (EM)	Regime-wtd AR
TAR	Classical reg.-switch	Threshold on y_{t-1}	Per-regime OLS
HMM + Ridge	Probabilistic ML	Baum-Welch EM	Per-regime Ridge
Mixture of Experts	Probabilistic ML	Soft gating via EM	Wtd Ridge
ML Regime (XGB)	Nonlinear ML	KMeans $\rightarrow$ XGBClassifier	Per-regime XGBRegressor
MSNN	Nonlinear ML	Markov chain + MLP (EM)	Per-regime MLP
Model Average	Ensemble	—	Equal-weight mean

Evaluation Metric — Lucas Sensitivity Ratio (LSR)

$$\mathrm{LSR} = \frac{\mathrm{RMSE}_{\mathrm{post}}}{\mathrm{RMSE}_{\mathrm{pre}}} \qquad \begin{cases} \mathrm{LSR} > 1 & \text{Lucas-critique vulnerable (post-break worse)} \\ \mathrm{LSR} < 1 & \text{inverse Lucas: variance-reducing break (post-break better)} \end{cases}$$

Model Details

HMM + Ridge

- Baum-Welch EM on $\mathbf{O}_t = [y_t,\, y_{t-1},\, \bar{y}^{(5)},\, s^{(5)},\, \bar{y}^{(20)}]^\top$
- Viterbi decoding $\rightarrow$ per-regime Ridge regressor
- Forward filter at test time — no look-ahead; soft posteriors adapt to shifted emissions without re-training

Mixture of Experts

- K=2 Ridge experts; logistic gating via EM
- Soft forecast: $\hat{y}_t = \sum_k \pi_k(x_t)\, \hat{\beta}_k^\top x_t$
- Partial regularisation; reduced sensitivity to regime misclassification

ML Regime (XGB)

- KMeans $\rightarrow$ XGBClassifier $\rightarrow$ per-regime XGBRegressor
- Frozen KMeans boundaries brittle to distributional shift
- Piecewise-constant leaves memorise pre-break structure $\rightarrow$ high LSR

MSNN

- Hamilton MSM with per-regime MLP experts ($n_{\text{feat}} \rightarrow 32 \rightarrow 16 \rightarrow 1$, ReLU, Adam)
- EM: forward-backward smoother $\rightarrow$ soft $\gamma_t(k)$; analytical $\hat{P}_{jk}$
- Markov chain + smoothed posteriors act as regularisers (analogous to HMM)

Results: RMSE Across All Experiments

	Simulated Mild			Simulated Severe			IP 1984			CPI 1979		
Model	Pre	Post	LSR	Pre	Post	LSR	Pre	Post	LSR	Pre	Post	LSR
AR(2)	1.634	2.706	1.66	1.634	4.261	2.61	0.717	0.514	0.72	2.631	2.594	0.99
ARMA(2,1)	1.674	5.577	3.33	1.674	11.61	6.94	0.811	0.514	0.63	2.533	3.556	1.40
MSM	1.673	5.968	3.57	1.673	11.96	7.15	0.823	0.723	0.88	2.954	4.038	1.37
TAR	1.537	2.526	1.64	1.537	4.266	2.78	0.624	0.397	0.64	1.912	2.276	1.19
HMM	1.696	2.559	**1.51**	1.696	3.892	**2.29**	0.648	**0.378**	**0.58**	2.040	2.311	**1.13**
MoE	1.339	2.628	1.96	1.339	4.742	3.54	0.619	0.407	0.66	1.791	2.992	1.67
XGB	0.396	3.337	8.43	0.396	8.471	21.41	0.140	0.443	3.18	0.357	2.559	7.18
MSNN	0.613	**2.493**	4.07	0.613	**3.391**	5.53	0.326	0.493	1.51	1.852	3.068	1.66
Avg	1.222	2.015	2.47	1.222	6.007	4.92	0.521	0.401	0.77	1.757	2.268	1.29

LSR = post/pre RMSE. Green = best LSR. Red = worst. **Bold post** = best absolute post-break RMSE.

What Worked & Why

HMM — best LSR all 4 experiments

- *Flexible latent-state estimation* (soft posteriors), not functional form — probabilistic assignments adapt to shifted emission distributions without re-training.
- Ridge regressors resist overfitting to pre-break noise. LSR = 1.51 (mild), 2.29 (severe).

MSNN — best absolute post-break RMSE

- Smooth MLP extrapolation degrades more gracefully than XGB's step functions. Advantage grows with shift magnitude (2.493 mild, 3.391 severe).

Model average — stable ensemble

- Lower post-break RMSE *and* lower LSR in 2/4 experiments. Expanding $\mathcal{F}$ reduces approximation error (consistent with the risk decomposition).

XGB — catastrophic failure

- Piecewise-constant leaves memorise pre-break regime structure; post-break misclassification causes systematic leaf-value errors.
- Pre-break RMSE 0.40: a *denominator effect* inflating LSR to 8.43 (mild) / 21.41 (severe). Frozen KMeans regime classifier is most brittle under distributional shift.

MSM — correctly specified, then collapses

- Correctly-specified for the DGP. Fixed EM estimates $(\hat{\mu}_k, \hat{\sigma}_k, \hat{P})$ cause Hamilton filter collapse: regime accuracy 75%→20%, ARI→0.
- LSR = 3.57 — *worse* than AR(2) (1.66). Correct pre-break specification *increases* Lucas vulnerability.

Key Insight

Models flexible in *latent-state estimation* (HMM) outperform those flexible in *functional form* (XGB) under structural breaks.

Conclusions

Main Findings

1. **ML does not resolve the Lucas Critique** — better OOS forecasts can still worsen under unforeseen regime change
2. **Latent-state flexibility is key** — HMM best LSR in all 4 experiments; XGB worst despite lowest pre-break RMSE
3. **Correct specification can be detrimental** — MSM degrades more than AR(2) in all cases
4. **Break character matters** — variance-reducing breaks (IP 1984) produce an *inverse* Lucas Critique for 7/9 models
5. **LSR and post-break RMSE diverge** — both needed; MSNN and Model Average illustrate the trade-off

Policy Implication

ML does *not* make forecasting models policy-invariant. HMM-family models preferred where structural stability matters.

Future Work

- **Ensembling** — key insight: reduces approximation error; optimised stacking / BMA
- **Online updating** — where does model memory most improve resilience?
- Richer DGPs and more real-world series
- Attention-based sequence models